

On Asymptotic Properties of Verbal Subsets in a Group

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Abstract—We prove that the asymptotic density $\rho(w[F_r])$ of the verbal set $w[F_r]$ in a free group F_r , $r \geq 2$, is equal to 0 for every word w with exponent sum $\exp(w) \neq 1$. When $\exp(w) = 1$ one has $w[F_r] = F_r$, and so $\rho(w[F_r]) = 1$.

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1. INTRODUCTION

In this paper we describe asymptotic densities of verbal sets in free groups. In particular, we show that a proper verbal subset in a free non-abelian group is negligible, i.e., it has asymptotic density zero. This also follows from a much larger (and much harder) result on the description of definable sets in free groups [7], however, here we give a direct and simple proof.

1.1. Verbal Subsets

Let F be a free group of infinite countable rank with basis $X = \{x_1, \dots, x_k, \dots\}$. We view F as the set of all reduced words in X equipped with the standard multiplication (concatenation and consequent reduction). For a positive integer n by F_n we denote the free group of rank n with basis $X_n = \{x_1, \dots, x_n\}$, which is the subgroup of F generated by X_n . For an arbitrary group G and a word $w = w(x_1, \dots, x_n) \in F$ we denote by $w[G]$ the set of all images of w under all possible homomorphisms $\varphi : F \rightarrow G$. In other words, $w[G]$ consists of all elements of the form $w(g_1, \dots, g_n)$, where $g_1, \dots, g_n$ are elements of G . Let $w(G)$ be the verbal subgroup of G , generated by the set $w[G]$. A subset S of G is called *verbal* if $S = w[G]$ for some word $w \in F$.

In recent years there has been a great deal of interest in verbal sets and verbal subgroups in groups. We refer to [12, 9, 8, 17] for results in finite groups, to [15, 16, 18, 14, 10] for verbal width in infinite groups, and to [6, 5, 1, 2] for stable verbal length.

The structure of a verbal subset in F_n may well be very complex, as Razborov showed in [13] for the verbal sets defined by quadratic words of the type $\Pi_{i=1}^n [x_i, y_i]$. We proved in [11] that for any group word w and every $n \geq 2$ the set $w[F_n]$ is rational in F_n if and only if $w[F_n] = F_n$ or $w[F_n] = 1$. Recall, that a subset R of a monoid G is called *rational* if R can be obtained from the finite subsets of G by a sequence of *rational* operations, i.e., union, product and submonoid generation. Alternatively, a rational subset can be defined as a set recognized by a finite automaton over G . Furthermore, it was shown in [4] that proper verbal subsets in F_n are not even context-free.

For any positive integer $i \in \mathbb{N}$ and each word $w \in F$ we define by $\sigma_i(w)$ the *i-th exponent* of w that is equal to the sum of all exponents of x_i in w . By $\exp(w)$ we denote $\gcd(\sigma_i(w) | i \in \mathbb{N})$. We write $\exp(w) = 0$ when $\sigma_i(w) = 0$ for all $i \in \mathbb{N}$, and call such w a *commutator word*. In this case $w \in F'$, the derived (commutator) subgroup of F .

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For sake of completeness we formulate here the following known, easy, but useful result.

Lemma 1.1. *Let $w = w(x_1, \dots, x_n)$ be a word in F and G be an arbitrary group. Then $\exp(w) = 1$ implies $w[G] = G$ for an arbitrary group G . On the other hand, if $\exp(w) \neq 1$ then $w[F_r] \neq F_r$ for every $r \geq 2$.*

Proof. If $\exp(w) = 1$ then there are integers $k_1, \dots, k_n$ such that

$$k_1\sigma_1(w) + \dots + k_n\sigma_n(w) = 1.$$

Then for an arbitrary element g in G one has $g = w(g^{k_1}, \dots, g^{k_n})$, and the first claim follows. Otherwise, if $\exp(w) \neq 1$ then $w[A_r] \neq A_r$ even for a free abelian group A_r of rank r , hence the second claim. $\square$

Wicks found [21] an algorithm for solving quadratic equations of the form $[x, y] = g$ and $x^2y^2 = g$ in F_r , $r \in \mathbb{N}$. In particular, he proved that an element $g \neq 1$ in F_r is a commutator, i.e., the equation $[x_1, x_2] = g$ has a solution in F_r , if and only if g is a conjugate of a cyclically reduced word of the form $ABCA^{-1}B^{-1}C^{-1}$, with $A, B, C \in F_r$, and where at most one of A, B, C is equal to the identity.

We define the genus of an element $v \in F'$ to be the least positive integer c such that v is a product of c commutators in F . Every element of genus c in F' can be obtained by a non-cancelling substitution into one of finitely many fixed words, the so called *Wicks form* of genus c (see, for instance, [19, 20, 3]).

Given a reduced word $w = w(x_1, \dots, x_n)$ one may construct the verbal set $w[F_r]$ by repeating the follows: choose some reduced words $A_1, \dots, A_n \in F_r$, substitute them for variables $x_1, \dots, x_n$ in w , and then reduce obtaining a word in $w[F_r]$. Clearly, any two such words A_i and A_j may cancel at most once (if, as usual, we do the maximal possible cancellation) in the reduction process. Hence, there exists a finite number of all possible “cancellation schemes” in reduction of the words $w(A_1, \dots, A_n)$. It follows that for every word w there are finitely many words $u_i(y_1, \dots, y_m)$ such that every non-trivial word u from $w[F_r]$ can be presented without cancellation in the form $u_i(P_1, \dots, P_m)$ for some $P_1, \dots, P_m \in F_r$ and a suitable i . Notice, that the forms $u_i(y_1, \dots, y_m)$ are not quite the same as the Wick’s forms, since we do not claim that every reduced word of the form $u_i(P_1, \dots, P_m)$ actually gives an element from $w[F_r]$.

1.2. Asymptotic Densities

A *stratification* of a countable set T is a sequence $\{T_m\}_{m \in \mathbb{N}}$ of non-empty finite subsets T_m whose union is T . Stratifications are often specified by size functions. A *size* or *length function* on T is a map $\sigma : T \rightarrow \mathbb{N}$ from T to the nonnegative integers $\mathbb{N}$ such that the inverse image of every integer is finite. The corresponding spherical and ball stratifications are formed by *spheres* $\mathbb{S}_m = \{t \in T \mid \sigma(t) = m\}$ and *balls* $\mathbb{B}_m = \{t \in T \mid \sigma(t) \leq m\}$.

The *asymptotic density* of $A \subset T$ with respect to a stratification $\{T_m\}$ is defined to be

$$\rho(A) = \lim_{m \rightarrow \infty} \rho_m(A), \quad (1)$$

where $\rho_m(A) = \frac{|A \cap T_m|}{|T_m|}$, when the limit exists. Otherwise, we use the limits

$$\bar{\rho}(A) = \limsup_{m \rightarrow \infty} \rho_m(A), \quad \underline{\rho}(A) = \liminf_{m \rightarrow \infty} \rho_m(A). \quad (2)$$

Note, that some authors call $\rho(A)$ the strict asymptotic density and refer to $\bar{\rho}$ as the standard asymptotic density.

A subset $A \subseteq T$ is said to be *generic* with respect to the stratification $\{T_m\}_{m \in \mathbb{N}}$ if $\rho(A) = 1$ and *negligible* if $\rho(A) = 0$. A property of elements of T is generic if it holds on a generic subset.

Similarly, one can define relative genericity of subsets in T . For subsets $A \subseteq B \subseteq T$ we define first a relative frequency $\rho_m(A|B)$ of A in B with respect to $\{T_m\}_{m \in \mathbb{N}}$ as the ratio

$$\rho_m(A|B) = \frac{|A \cap T_m|}{|B \cap T_m|} = \frac{\rho_m(A)}{\rho_m(B)}. \quad (3)$$

Here we treat the fractions $\frac{0}{0}$ as 0. In particular, $\rho_m(A|T) = \rho_m(A)$. Now the asymptotic density of A relative to B is defined by

$$\rho(A|B) = \lim_{m \rightarrow \infty} \rho_m(A|B). \quad (4)$$

We also consider

$$\bar{\rho}(A|B) = \limsup_{m \rightarrow \infty} \rho_m(A|B) \quad \text{and} \quad \underline{\rho}(A|B) = \liminf_{m \rightarrow \infty} \rho_m(A|B). \quad (5)$$

We say that A is generic in B relative to $\{T_m\}_{m \in \mathbb{N}}$ if $\rho(A|B) = 1$ and negligible if the limit is 0.

If the asymptotic density $\rho(A)$ does not exist we find ourselves in the context where the upper and lower densities $\bar{\rho}$ and $\underline{\rho}$ could be used as a coarse approximation to ρ . In this case we say that A has an *intermediate asymptotic density* relative to B if $1 > \bar{\rho}(A|B) \geq \underline{\rho}(A|B) > 0$.

2. ASYMPTOTIC DENSITY OF VERBAL SETS IN A FREE GROUP

Let F_r be a free group and $w = w(x_1, \dots, x_n)$ be an arbitrary reduced word. An equation

$$w = g, \quad \text{where} \quad g \in F_r, \quad (6)$$

is solvable in F_r if and only if g belongs to $w[F_r]$. Hence, the question on the “probability” of the equation (6) to have a solution in F_r reduces to the question on the asymptotic density of the verbal set $w[F_r]$ in F_r . By Lemma 1.1 if $\exp(w) = 1$ then $\rho(w[F_r]) = 1$ for every stratification $\{T_m\}_{m \in \mathbb{N}}$ of F_r .

It is customary in group theory and combinatorics to consider the word length $|\cdot| : F_r \rightarrow \mathbb{N}$ in the basis X_r as the size function on F_r . The corresponding stratifications $\{\mathbb{S}_m\}_{m \in \mathbb{N}}$ and $\{\mathbb{B}_m\}_{m \in \mathbb{N}}$ are the standard spherical and ball stratifications on F_r .

2.1. Asymptotic Densities of Verbal Sets in the Infinite Cyclic Group

It is easy to compute the asymptotic density of verbal sets of the infinite cyclic group F_1 .

Proposition 2.1. The following holds in the infinite cyclic group F_1 with respect to the spherical stratification $\{\mathbb{S}_m\}_{m \in \mathbb{N}}$:

- if $\exp(w) = 1$ then $\rho(w[F_1]) = 1$, thus $w[F_1]$ is generic;
- if $\exp(w) = 0$, i.e. w is commutator word, then $\rho(w[F_1]) = 0$, thus $w[F_1]$ is negligible;
- if $\exp(w) = k$, where $k \neq 0, 1$, then $\bar{\rho}(w[F_1]) = 1$ and $\underline{\rho}(w[F_1]) = 0$.

Proof. The first statement follows from Lemma 1.1 and the second one is obvious. The third statement is also clear because $w[F_1] \cap \mathbb{S}_m = \mathbb{S}_m$ when $k|m$, and $w[F_1] \cap \mathbb{S}_m = \emptyset$, when $k \nmid m$. $\square$

Proposition 2.2. The following holds in the infinite cyclic group F_1 with respect to the ball stratification $\{\mathbb{B}_m\}_{m \in \mathbb{N}}$:

- if $\exp(w) = 1$ then $\rho(w[F_1]) = 1$, thus $w[F_1]$ is generic;
- if $\exp(w) = 0$, i.e., w is a commutator word, then $\rho(w[F_1]) = 0$, thus $w[F_1]$ is negligible;
- if $\exp(w) = k$, where $k \neq 0, 1$, then $\rho(w[F_1]) = 1/k$.

Proof. The first two statements follow by the same argument as in Proposition 2.1. The third one can be derived by a direct computation

$$\frac{|w[F_1] \cap \mathbb{B}_m|}{|\mathbb{B}_m|} = \frac{2[m/k] + 1}{2m + 1} \rightarrow_{k \rightarrow \infty} 1/k. \quad (7)$$

$\square$

2.2. Asymptotic Densities of Verbal Sets in a Non-Cyclic Free Group

Theorem 2.3. *Let $F_r, r \geq 2$, be a non-cyclic free group with basis $X_r = \{x_1, \dots, x_r\}$. Then the following holds with respect to the spherical, as well as the ball, stratifications of F_r :*

- if $\exp(w) = 1$ then $\rho(w[F_r]) = 1$, thus $w[F_r]$ is generic;
- if $\exp(w) \neq 1$, then $\rho(w[F_r]) = 0$, thus $w[F_r]$ is negligible.

Proof. The first statement follows by Lemma 1.1.

To prove the second statement we assume that $\exp(w) = 0$ or $\exp(w) = e \neq 1$. As was described in the introduction every element in the intersection $w[F_r] \cap \mathbb{S}_k$ is the image of a non-cancelling substitution of some $P_1, \dots, P_m \in F_r$ into one of the finitely many cancellation schemes $u(y_1, \dots, y_m)$ with $|u(P_1, \dots, P_m)| = k$. In a natural way we define the exponent sum $\tau_i(u)$ as the sum of the exponents of all occurrences of the piece P_i in u for all $i = 1, \dots, m$. We claim that $\tau_i(u) = 0$ or is divided by e . The latter is possible only if $e \neq 0$. Indeed, similar assertion is obviously true when we consider the result of first substitution $w(A_1, \dots, A_n)$ (see the description of the process in the introduction). Then we follow the reductio process and along the way we introduce the pieces P_i for $i = 1, \dots, m$. Every piece P occurs as the maximal cancellation of two words A_t, A_s or as a “middle” of some A_t consisting of those letters that do not cancel at all. Hence the exponent sum w.r.t. P in the form u obtained as the result of reduction of $w(A_1, \dots, A_n)$ is equal to the algebraic sums of corresponding exponents in the previous words. Hence, this sum is either equal to 0 or is a multiple of e . We can conclude that every piece P_i occurs in u at least two times (may be with different exponents ± 1). A value of a non-cancelling substitution to u of the pieces P_i for $i = 1, \dots, m$ depends on not more than $k/2$ letters. A number of such substitutions is $\leq (2r)^{k/2}$. Since $|\mathbb{S}_k| = 2r(2r-1)^{k-1}$ we derive that the asymptotic density for every cancellation scheme u is 0. Since there are only finitely many cancellation schemes u the second claim in the case of spheres follows. The result on the ball stratification is obtained in a similar way. $\square$

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